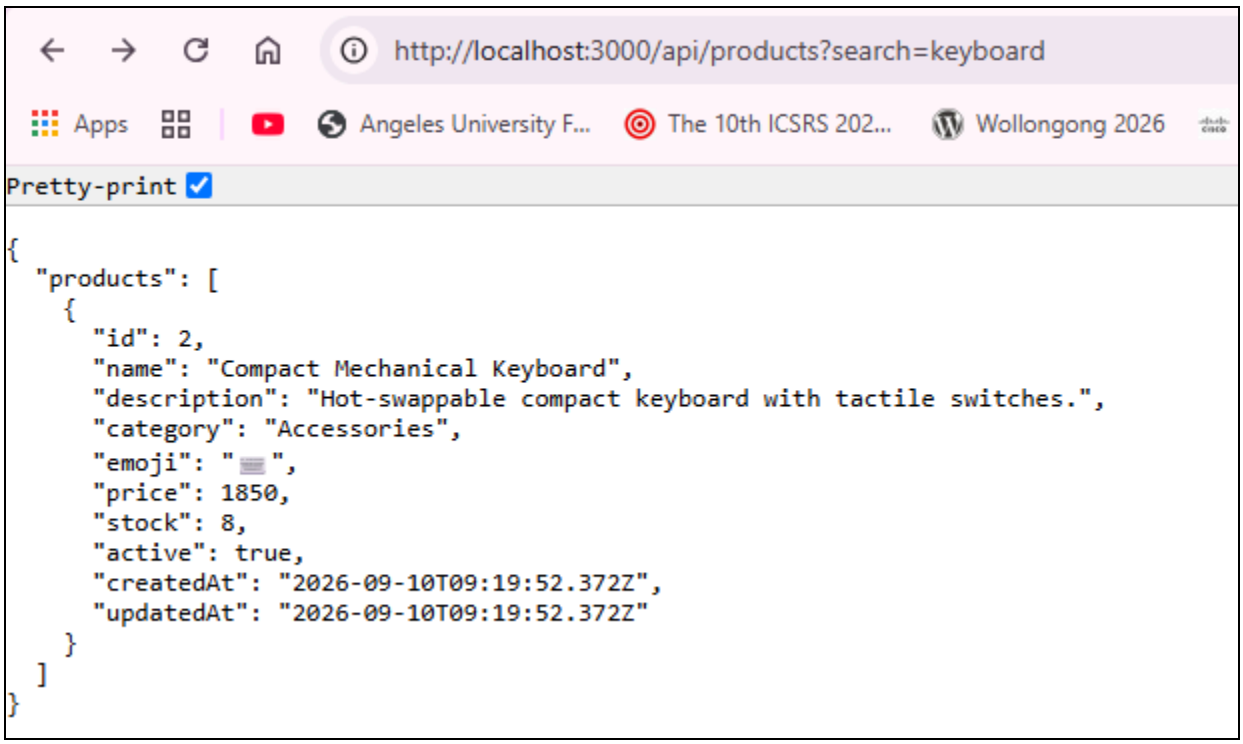
ANGELES UNIVERSITY FOUNDATION
Angeles City
COLLEGE OF COMPUTER STUDIES
1st Semester, S.Y. 2026 - 2027

PDC50: Full-Stack AI Application Development
ShopSmart Live Catalog Search Through the API

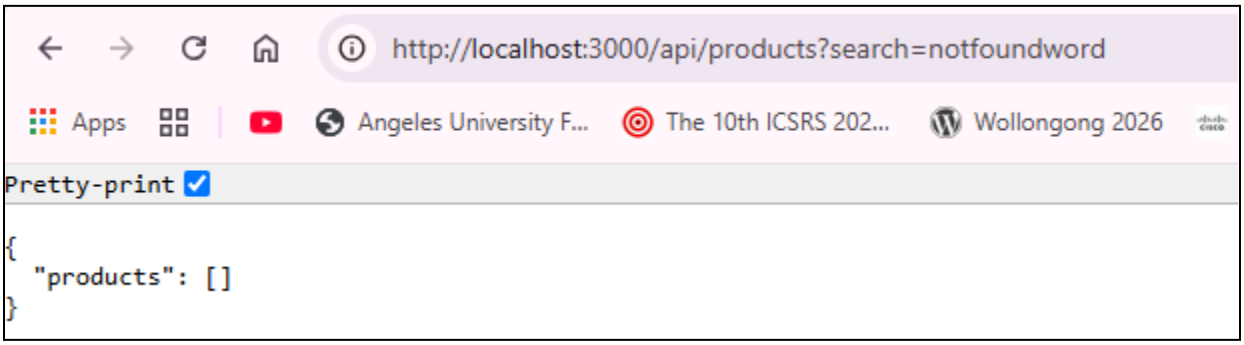
Student Name: Sean Gabriel S. Santos	**Submitted to:** 21-0207-737
Course, Year and Section: BS Information Technology 4	**Date Submitted:** October 16, 2026

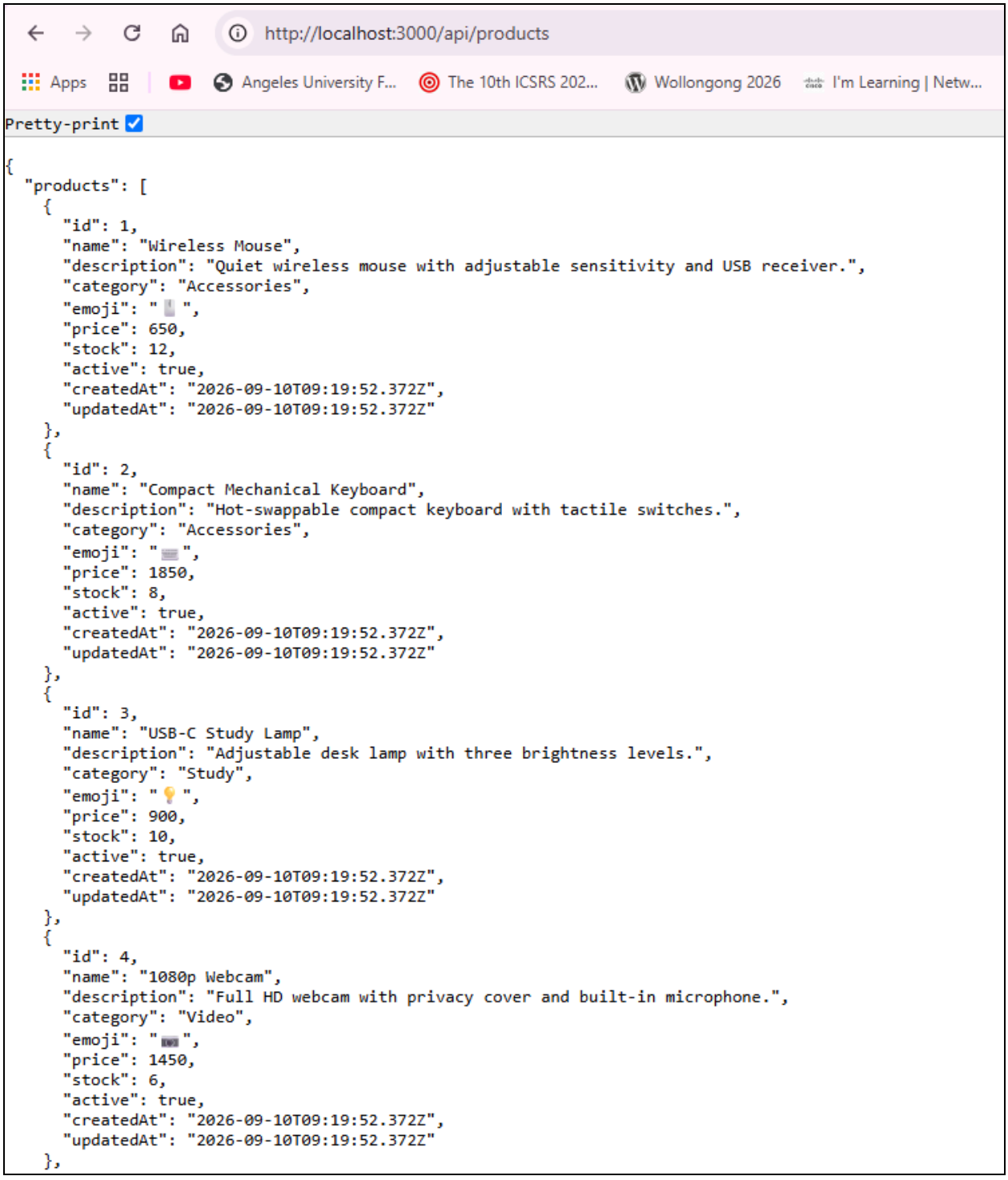
CHALLENGE 1: Improve Search on the Server



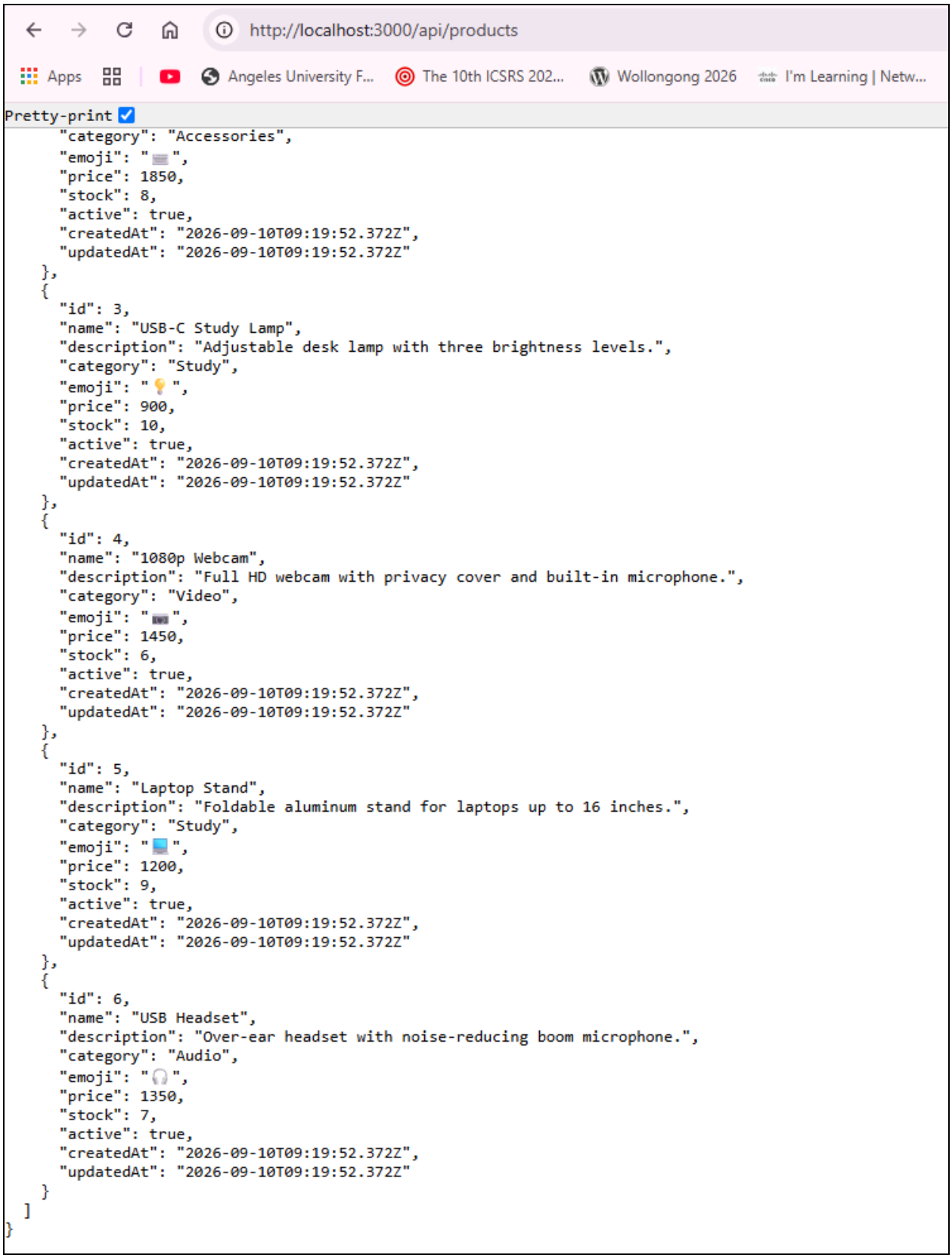

```
{
  "products": [
    {
      "id": 2,
      "name": "Compact Mechanical Keyboard",
      "description": "Hot-swappable compact keyboard with tactile switches.",
      "category": "Accessories",
      "emoji": "⌨",
      "price": 1850,
      "stock": 8,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    }
  ]
}
```


http://localhost:3000/api/products?search=privacy
Apps
Angeles University F...
The 10th ICSRS 202...
Wollongong 2026
Pretty-print
{
"products": [
{
"id": 4,
"name": "1080p Webcam",
"description": "Full HD webcam with privacy cover and built-in microphone.",
"category": "Video",
"emoji": "
",
"price": 1450,
"stock": 6,
"active": true,
"createdAt": "2026-09-10T09:19:52.372Z",
"updatedAt": "2026-09-10T09:19:52.372Z"
}
]
}


http://localhost:3000/api/products?search=notfoundword
Apps
Angeles University F...
The 10th ICSRS 202...
Wollongong 2026
Pretty-print
{
"products": []
}

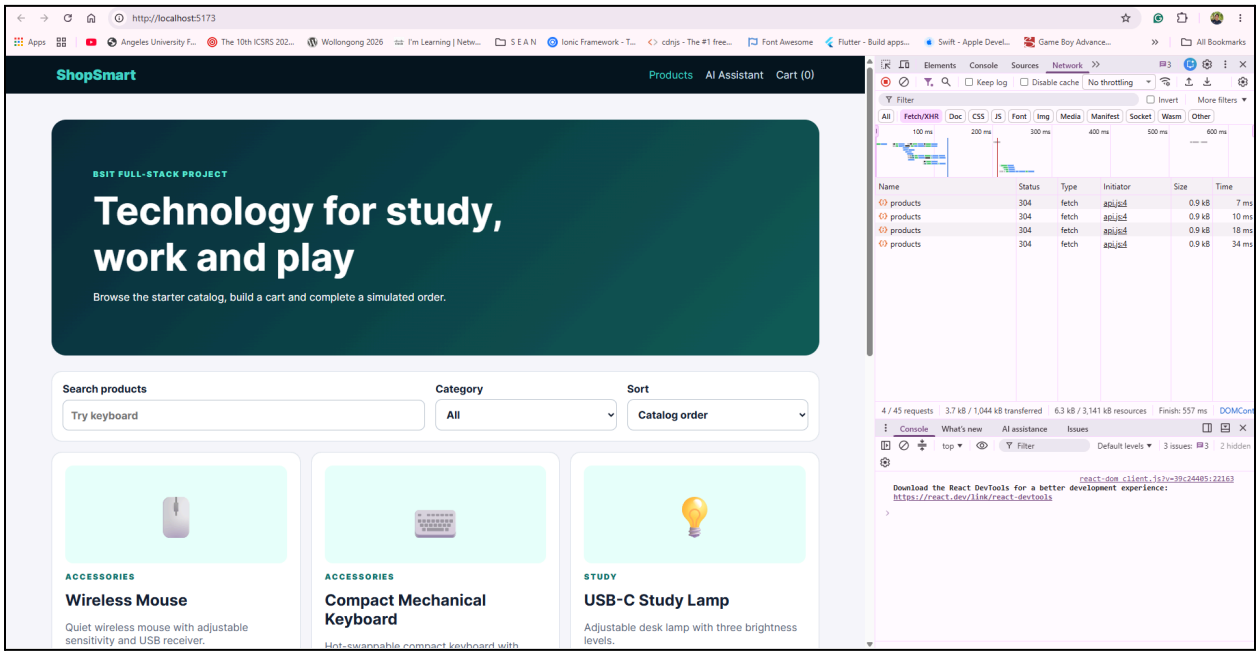


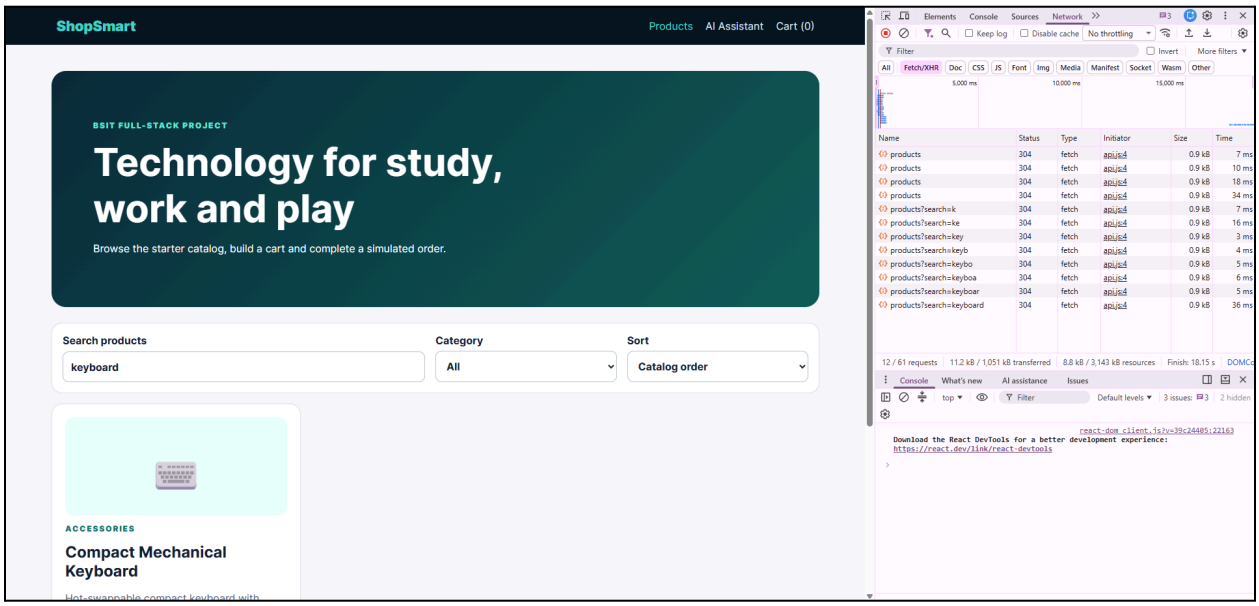

```
{
  "products": [
    {
      "id": 1,
      "name": "Wireless Mouse",
      "description": "Quiet wireless mouse with adjustable sensitivity and USB receiver.",
      "category": "Accessories",
      "emoji": "🖱️",
      "price": 650,
      "stock": 12,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
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    },
    {
      "id": 2,
      "name": "Compact Mechanical Keyboard",
      "description": "Hot-swappable compact keyboard with tactile switches.",
      "category": "Accessories",
      "emoji": "⌨️",
      "price": 1850,
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      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
    {
      "id": 3,
      "name": "USB-C Study Lamp",
      "description": "Adjustable desk lamp with three brightness levels.",
      "category": "Study",
      "emoji": "💡",
      "price": 900,
      "stock": 10,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
    {
      "id": 4,
      "name": "1080p Webcam",
      "description": "Full HD webcam with privacy cover and built-in microphone.",
      "category": "Video",
      "emoji": "📷",
      "price": 1450,
      "stock": 6,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
```

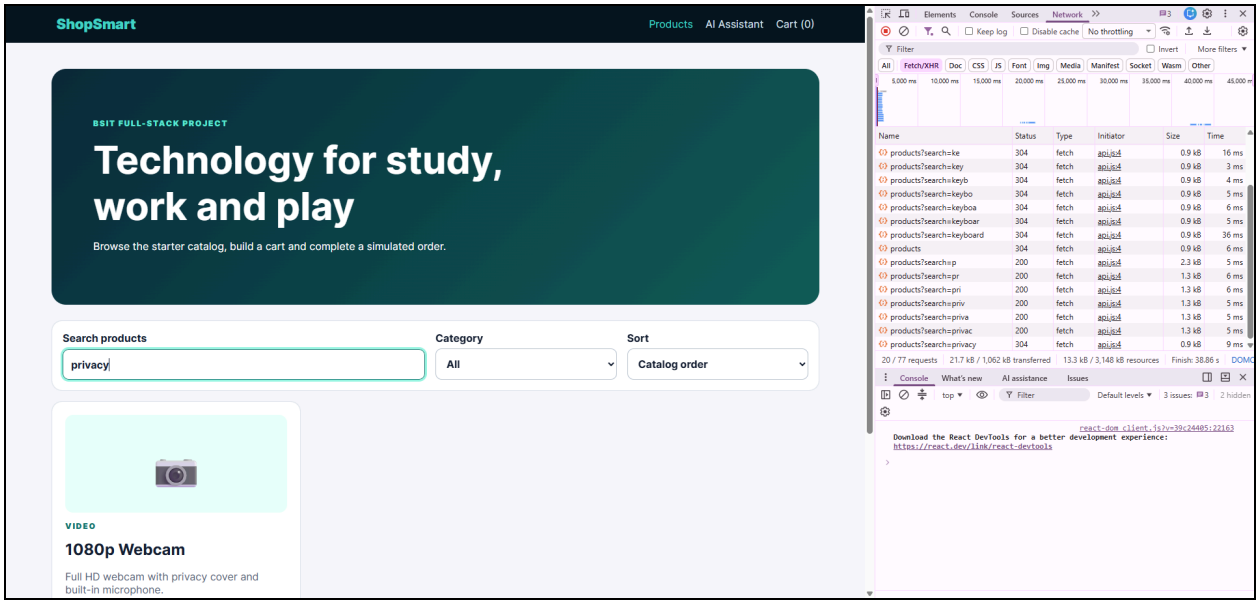


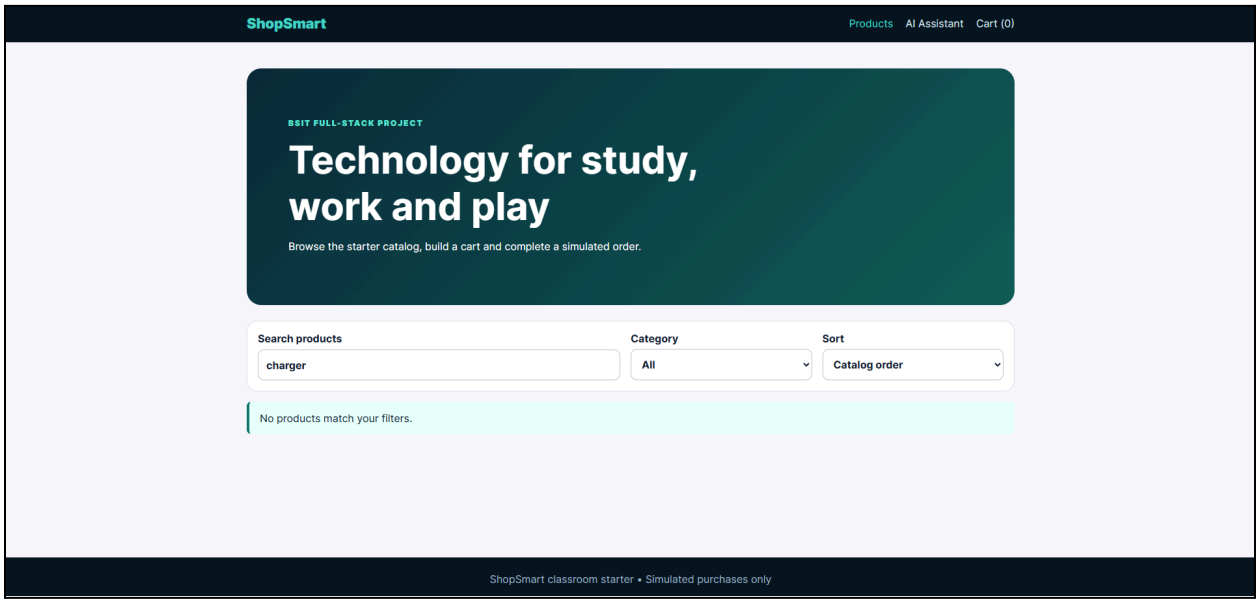

```
      "category": "Accessories",
      "emoji": "⌨️",
      "price": 1850,
      "stock": 8,
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      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
    {
      "id": 3,
      "name": "USB-C Study Lamp",
      "description": "Adjustable desk lamp with three brightness levels.",
      "category": "Study",
      "emoji": "💡",
      "price": 900,
      "stock": 10,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
    {
      "id": 4,
      "name": "1080p Webcam",
      "description": "Full HD webcam with privacy cover and built-in microphone.",
      "category": "Video",
      "emoji": "📹",
      "price": 1450,
      "stock": 6,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
    {
      "id": 5,
      "name": "Laptop Stand",
      "description": "Foldable aluminum stand for laptops up to 16 inches.",
      "category": "Study",
      "emoji": "💻",
      "price": 1200,
      "stock": 9,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
      "updatedAt": "2026-09-10T09:19:52.372Z"
    },
    {
      "id": 6,
      "name": "USB Headset",
      "description": "Over-ear headset with noise-reducing boom microphone.",
      "category": "Audio",
      "emoji": "🎧",
      "price": 1350,
      "stock": 7,
      "active": true,
      "createdAt": "2026-09-10T09:19:52.372Z",
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    }
  ]
}
```

CHALLENGE 2: Connect the Page to the API Search

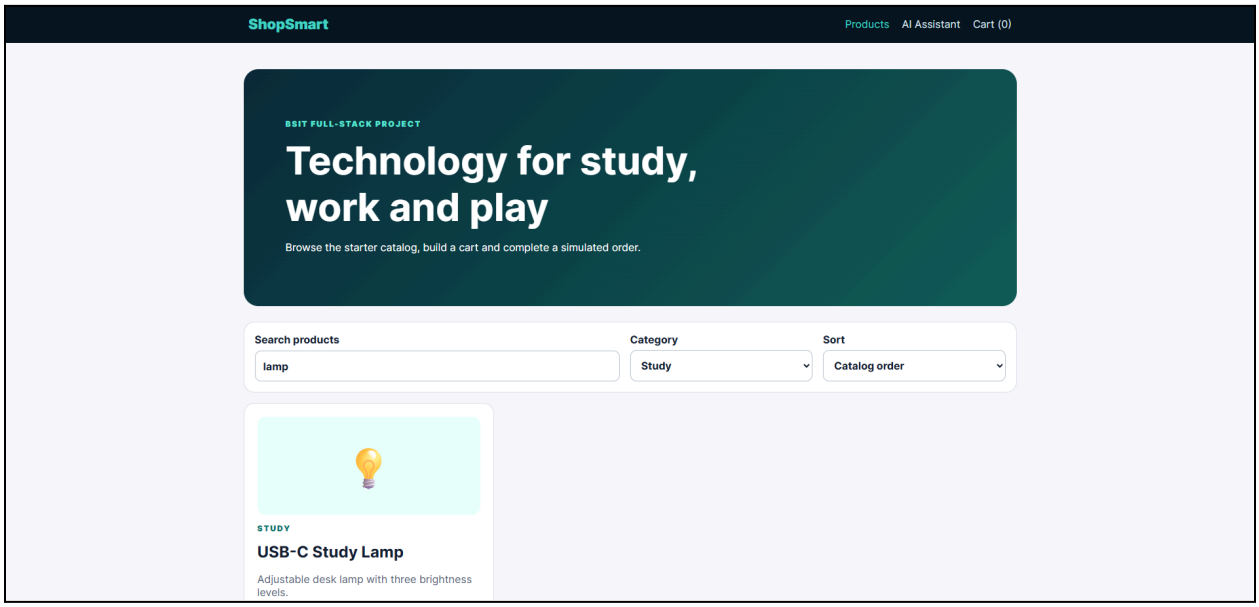


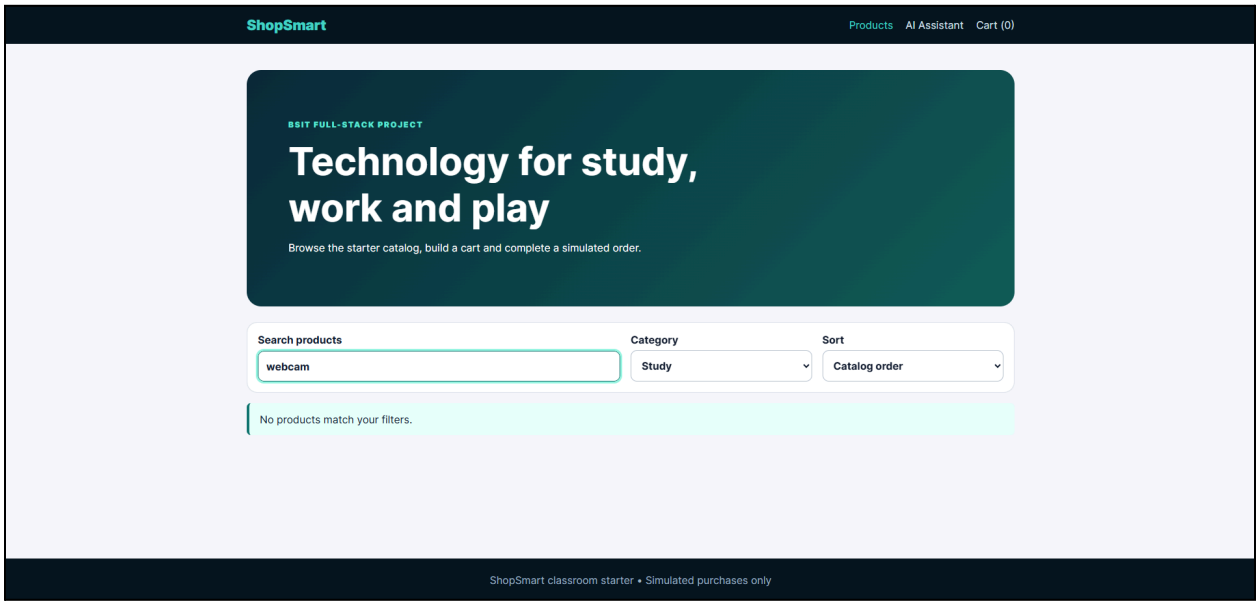


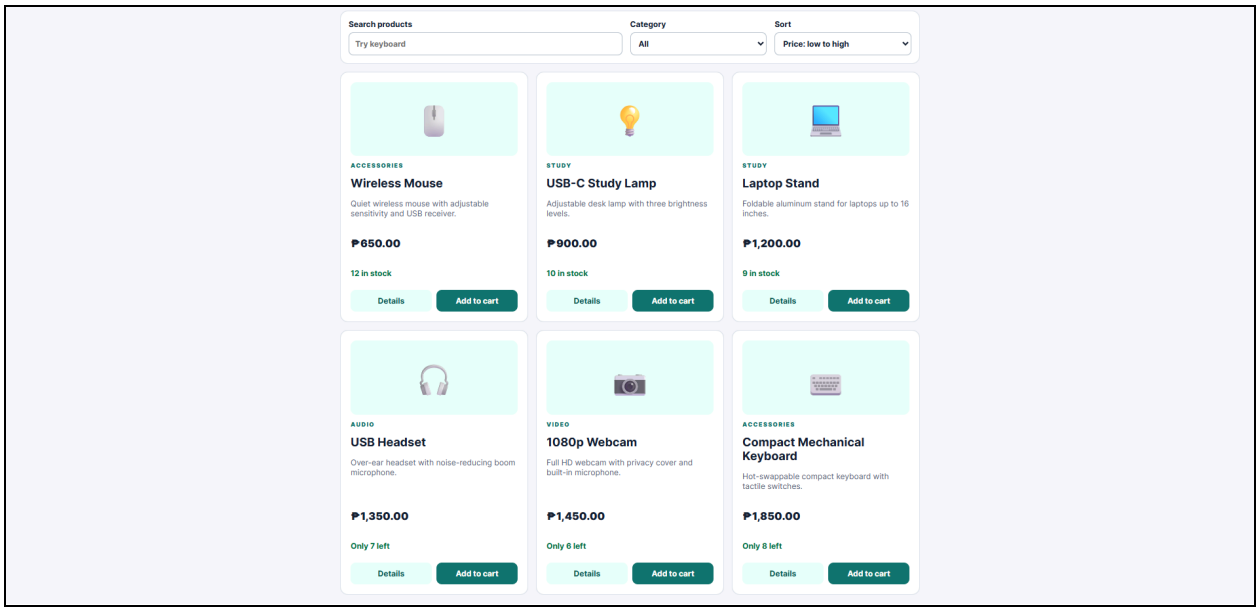

ShopSmart
Products
AI Assistant
Cart (0)
BSIT FULL-STACK PROJECT
Technology for study, work and play
Browse the starter catalog, build a cart and complete a simulated order.
Search products
privacy
Category
All
Sort
Catalog order
VIDEO
1080p Webcam
Full HD webcam with privacy cover and built-in microphone.
Elements
Console
Sources
Network
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools


ShopSmart
Products
AI Assistant
Cart (0)
BSIT FULL-STACK PROJECT
Technology for study, work and play
Browse the starter catalog, build a cart and complete a simulated order.
Search products
charger
Category
All
Sort
Catalog order
No products match your filters.
ShopSmart classroom starter • Simulated purchases only

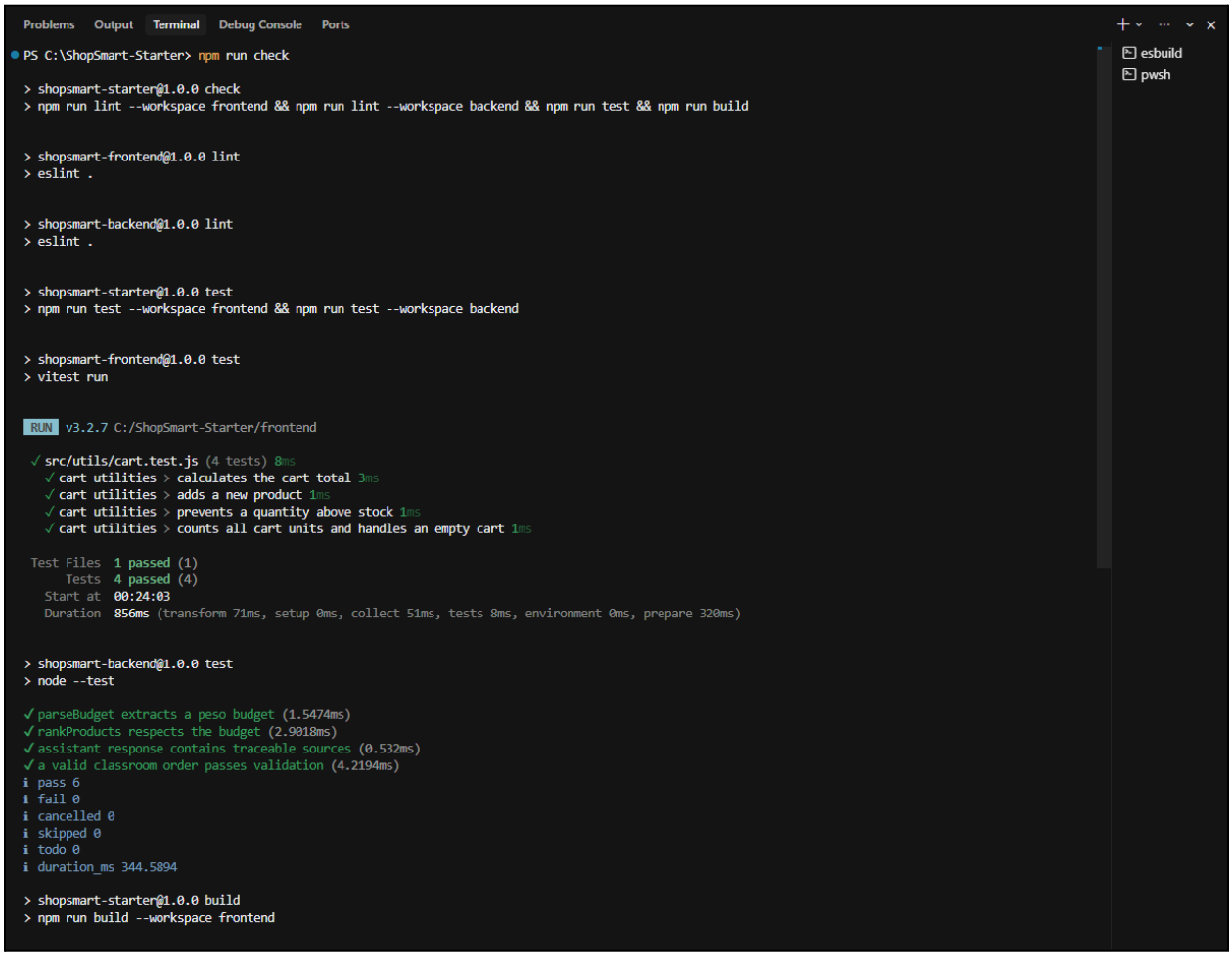
CHALLENGE 3: Preserve the Earlier Catalog Controls

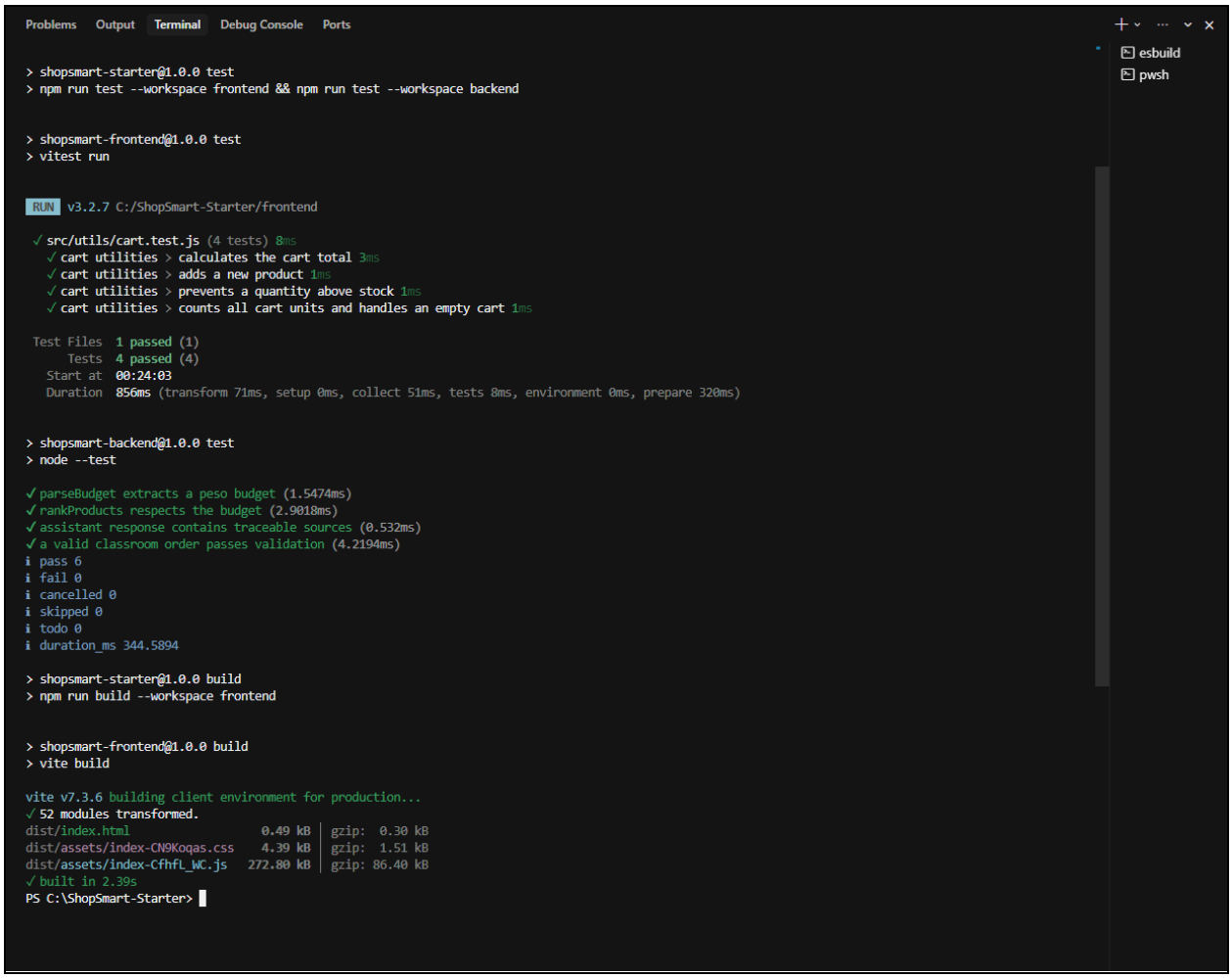






Final “npm run check” Output





When a shopper types in the search bar, the products page saves that text and an effect calls api.getProducts with the trimmed term. That function sends a GET request to /api/products, and it adds ?search= when the box has text. Express reads the query, trims it and asks Prisma for rows in PostgreSQL. The server keeps only active products and returns them in id order. If a term is present, a product matches when that term appears in the name or in the description and capital letters do not change the match. If the shopper types again before the first answer arrives, the effect marks the older request so its JSON cannot replace the list. Category names come from a separate request for the full catalog when the page opens, so the dropdown stays complete even when the visible products shrink to one item or to none. The page then filters that latest server list by the chosen category and sorts it in the browser.